

Data Research:World Doping Analysis

2026-06-28

R Markdown

Project 1 Story

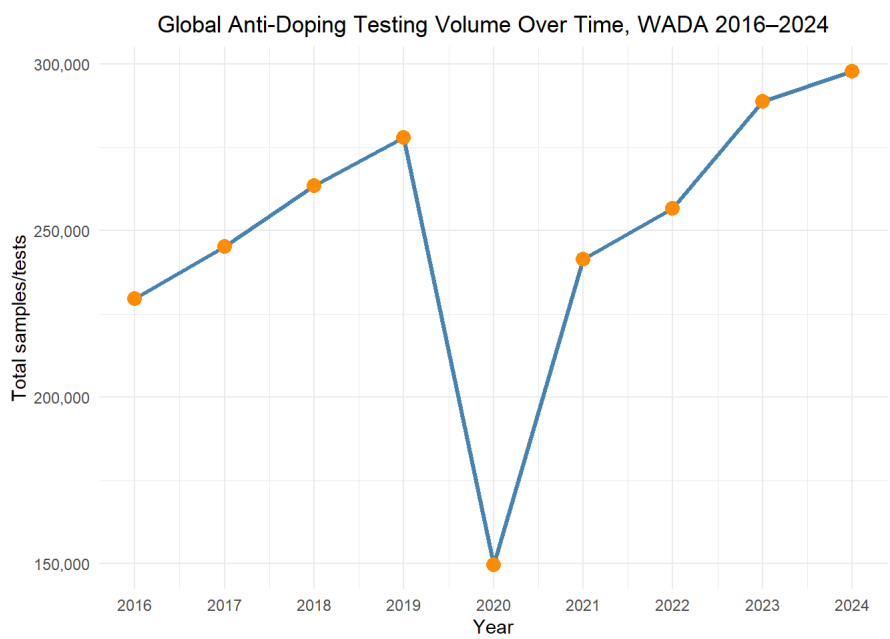
- 1. Testing volume “Which sports receive the most testing?”
- 2. AAF risk “Which sports show more suspicious findings?”
- 3. Mismatch “Are the most tested sports also the sports with the strongest detected-risk signals?”
- 4. ADRV confirmation “Do suspicious findings become confirmed violations?”
- 5. Final global risk interpretation “Which sports appear aligned, over-tested, or potentially under-tested?”

Research Question 1: How have global anti-doping testing volumes changed across sports from 2016 to 2024?

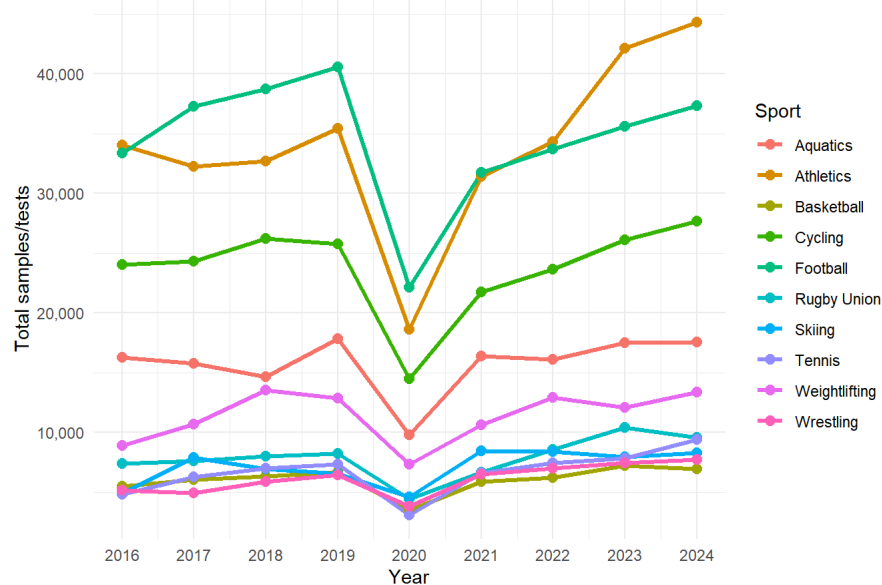
The first research question examines how global anti-doping testing volumes changed over time. Instead of only looking at the total number of tests per year, this section focuses on the **top 10 most tested sports worldwide** between 2016 and 2024.

This approach is useful because global testing is not distributed equally across all sports. A small number of sports receive a large share of testing resources, so comparing their yearly trends helps show which sports drive the overall global testing pattern.

```
## # A tibble: 9 × 5
##   year total_samples total_aafs sports_count aaf_rate_percent
##   <dbl>         <dbl>         <dbl>         <int>         <dbl>
## 1  2016           229514           3032           165           1.32
## 2  2017           245232           2749           170           1.12
## 3  2018           263519           2771           181           1.05
## 4  2019           278047           2702           175           0.972
## 5  2020           149758           1009           143           0.674
## 6  2021           241430           1560           127           0.646
## 7  2022           256769           1986           144           0.773
## 8  2023           288865           2313           155           0.801
## 9  2024           297965           2310           156           0.775
```



Testing Volume Trends for the Top 10 Most Tested Sports, WADA 2016–2024



Observation

The chart shows that global anti-doping testing is concentrated in a small group of sports. Football, Athletics, and Cycling remain among the most heavily tested sports across the period, although their yearly testing volumes follow different patterns.

A clear decline appears in 2020 across almost all of the top sports. This suggests that the COVID-19 period strongly disrupted global anti-doping testing activity. After 2020, testing volumes generally recovered, with Athletics showing a particularly strong increase after 2021 and reaching one of the highest testing levels by 2023 and 2024.

Overall, this analysis shows that global testing priorities are not evenly distributed across sports. A small number of sports receive a large share of testing resources, which provides the foundation for the next research question: whether these testing priorities match detected doping-risk signals such as AAF rates and ADRV outcomes.

Research Question 2: Which sports show the strongest AAF-based doping-risk signals?

After examining testing volume, the next step is to examine detected doping-risk signals. In this section, the focus is on **Adverse Analytical Findings (AAFs)**. An AAF does not necessarily mean a confirmed doping violation, but it is an important early indicator of detected risk in anti-doping testing.

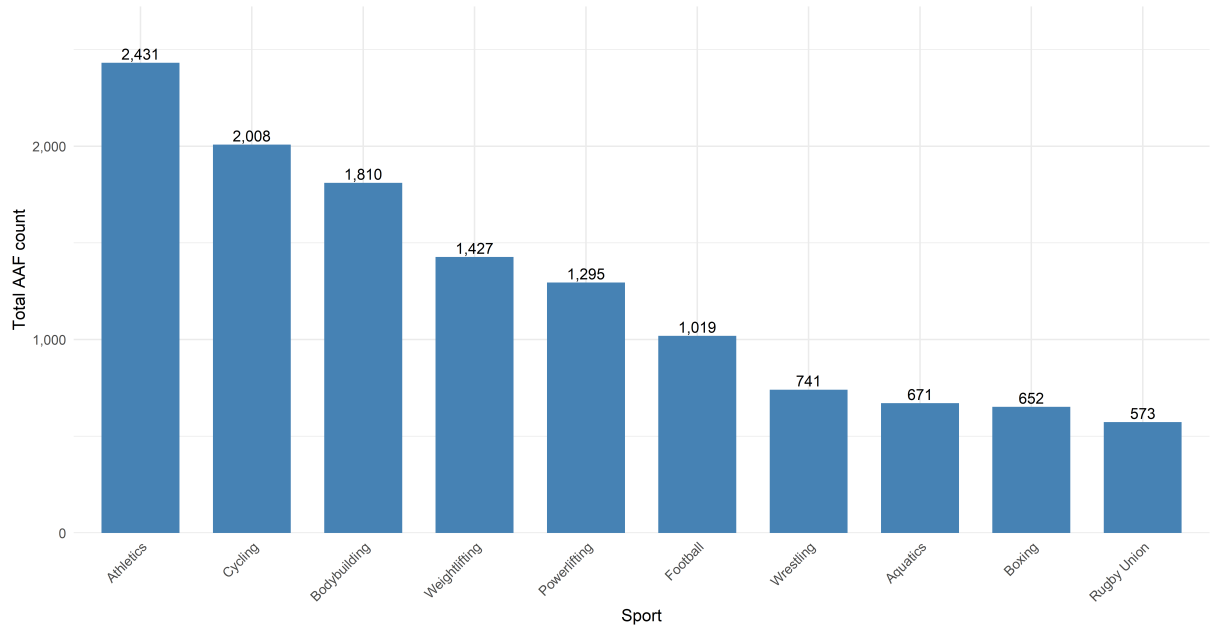
This research question considers two perspectives. First, sports are ranked by the total number of AAFs. This shows where the largest number of suspicious findings occurred. Second, sports are ranked by AAF rate per 1,000 tests. This adjusts for testing volume and therefore gives a better indication of relative detected risk.

To avoid misleading results from very small sports with only a few tests, a minimum testing threshold is applied before ranking sports by AAF rate.

2.1 Sports with the highest number of AAFs

```
## # A tibble: 10 × 5
##   sport_clean total_samples_2016_2024 total_aafs_2016_2024
##   <chr>          <dbl>          <dbl>
## 1 Athletics      305256          2431
## 2 Cycling        214017          2008
## 3 Bodybuilding   11200           1810
## 4 Weightlifting  102298          1427
## 5 Powerlifting    31168           1295
## 6 Football       310424          1019
## 7 Wrestling      54893            741
## 8 Aquatics       141864            671
## 9 Boxing         43031            652
## 10 Rugby Union    70806            573
## # 2 more variables: aaf_rate_per_1000_tests <dbl>, aaf_share_percent <dbl>
```

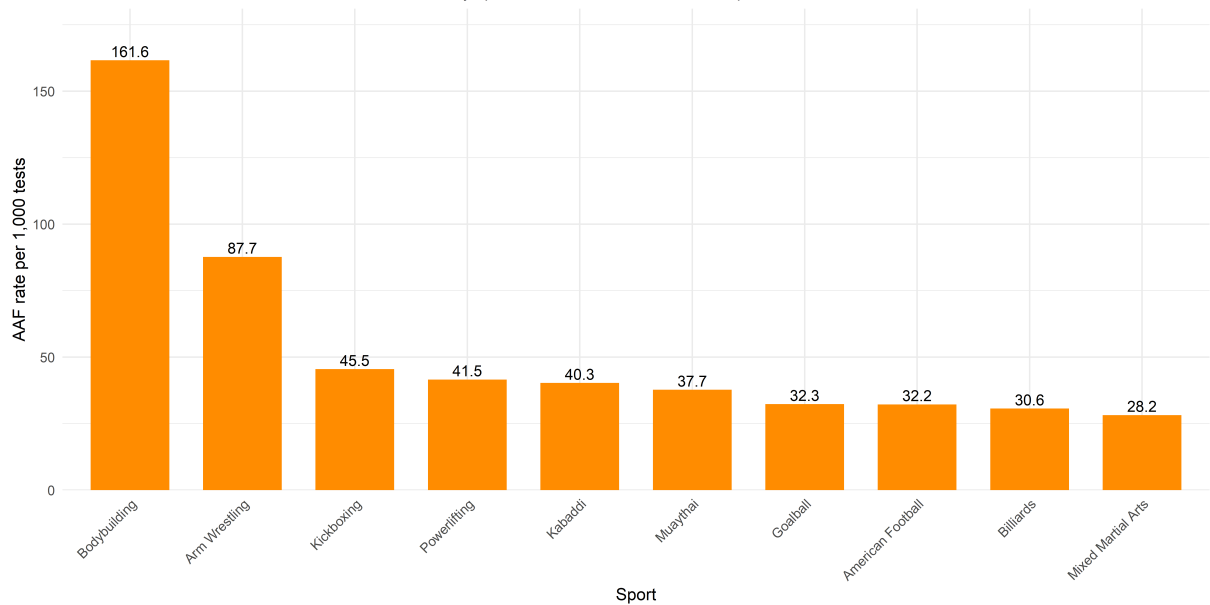
Top 10 Sports by AAF Count, WADA 2016–2024



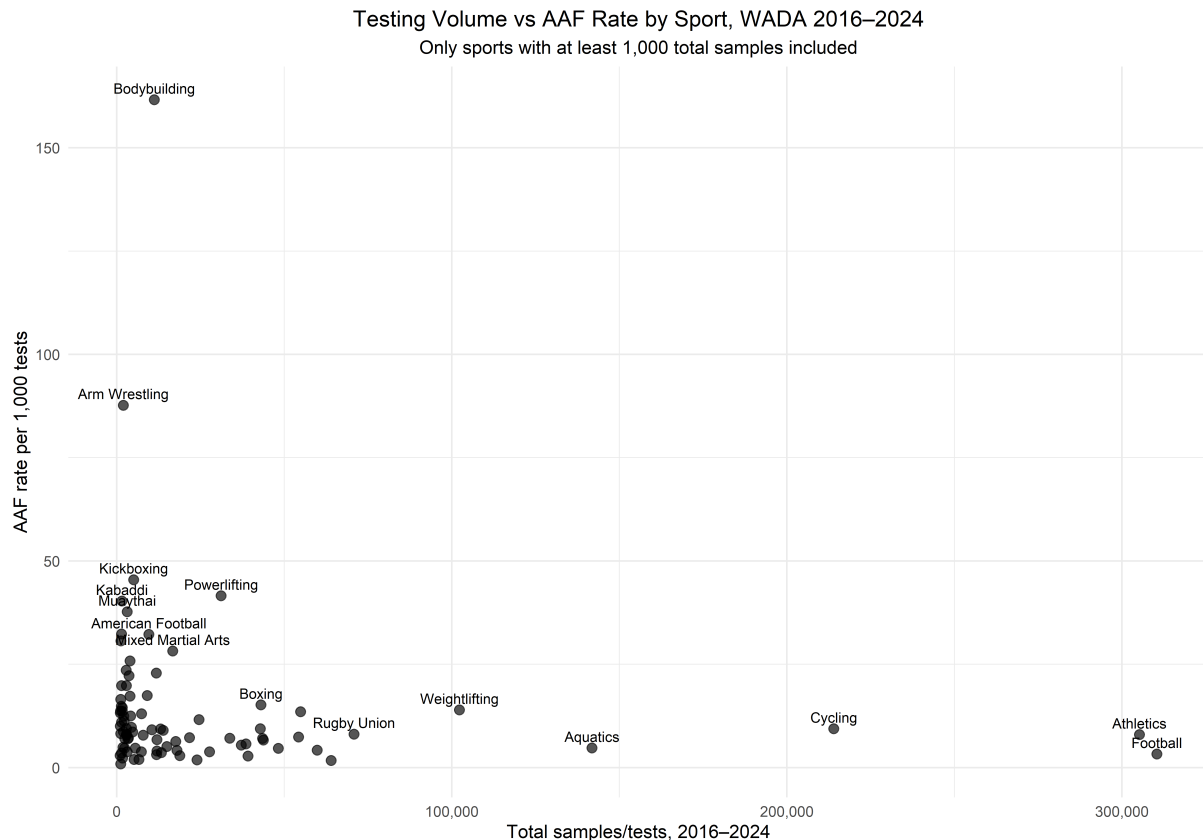
2.2 Sports with the highest AAF rate

```
## # A tibble: 10 × 4
##   sport_clean      total_samples_2016_2024 total_aafs_2016_2024
##   <chr>                <dbl>                <dbl>
## 1 Bodybuilding          11200                1810
## 2 Arm Wrestling          1973                 173
## 3 Kickboxing             5037                 229
## 4 Powerlifting          31168                1295
## 5 Kabaddi                1564                  63
## 6 Muaythai              3157                 119
## 7 Goalball              1423                  46
## 8 American Football     9596                 309
## 9 Billiards             1240                  38
## 10 Mixed Martial Arts   16742                 472
## # i 1 more variable: aaf_rate_per_1000_tests <dbl>
```

Top 10 Sports by AAF Rate, WADA 2016–2024
Only sports with at least 1,000 total samples included



2.3 Relationship between testing volume and AAF rate



Observation

The AAF analysis shows that the interpretation depends strongly on whether we look at total AAF counts or AAF rates. Sports with very large testing volumes, such as Athletics, Cycling, Football, Weightlifting, and Aquatics, naturally appear among the highest sports by total AAF count because they receive many tests.

However, the AAF rate per 1,000 tests gives a different view. After applying a minimum testing threshold, sports such as Bodybuilding, Arm Wrestling, Kickboxing, Powerlifting, Kabaddi, Muaythai, American Football, and Mixed Martial Arts show much higher relative AAF rates. This suggests that some sports with lower overall testing volume may show stronger detected-risk signals once testing volume is taken into account.

The scatter plot supports this point visually. Some highly tested sports have moderate or low AAF rates, while some less-tested sports have much higher AAF rates. Therefore, testing volume alone does not fully represent detected doping risk. This finding leads directly into the next research question, where testing share will be compared with AAF share to identify possible over-tested and under-tested sports.

This Research Question is stronger than just plotting “top 10 AAF rate” because it gives three layers:

AAF count = where most findings occurred. AAF rate = where findings were most frequent relative to testing volume. Scatter plot = whether testing volume and AAF risk seem aligned.

After this, RQ3(Research Question 3) can use testing_share, aaf_share, and mismatch_score to directly answer the testing-risk mismatch question.

Research Question 3: Which sports appear over-tested or under-tested based on testing share and AAF share? or Research Question 3: Are global testing priorities aligned with AAF-based detected risk?

After identifying the most tested sports and the sports with the strongest AAF-based risk signals, this section compares both dimensions directly. The aim is to evaluate whether global anti-doping testing volume appears aligned with detected doping-risk signals.

For each sport, two shares are calculated:

- **Testing share:** the sport's proportion of all global tests.
- **AAF share:** the sport's proportion of all global Adverse Analytical Findings.

The difference between these two values is used as a **Testing–Risk Mismatch Score**:

$$\text{Mismatch Score} = \text{AAF Share} - \text{Testing Share}$$

A positive mismatch score suggests that a sport produced more AAFs than expected based on its testing share. A negative mismatch score suggests that a sport received a relatively large share of testing but produced fewer AAFs relative to that testing volume.

3.1 Sports with the largest positive mismatch score

A positive mismatch score means that a sport's share of AAFs is larger than its share of testing. These sports may represent possible under-tested or higher-yield sports from an AAF-risk perspective.

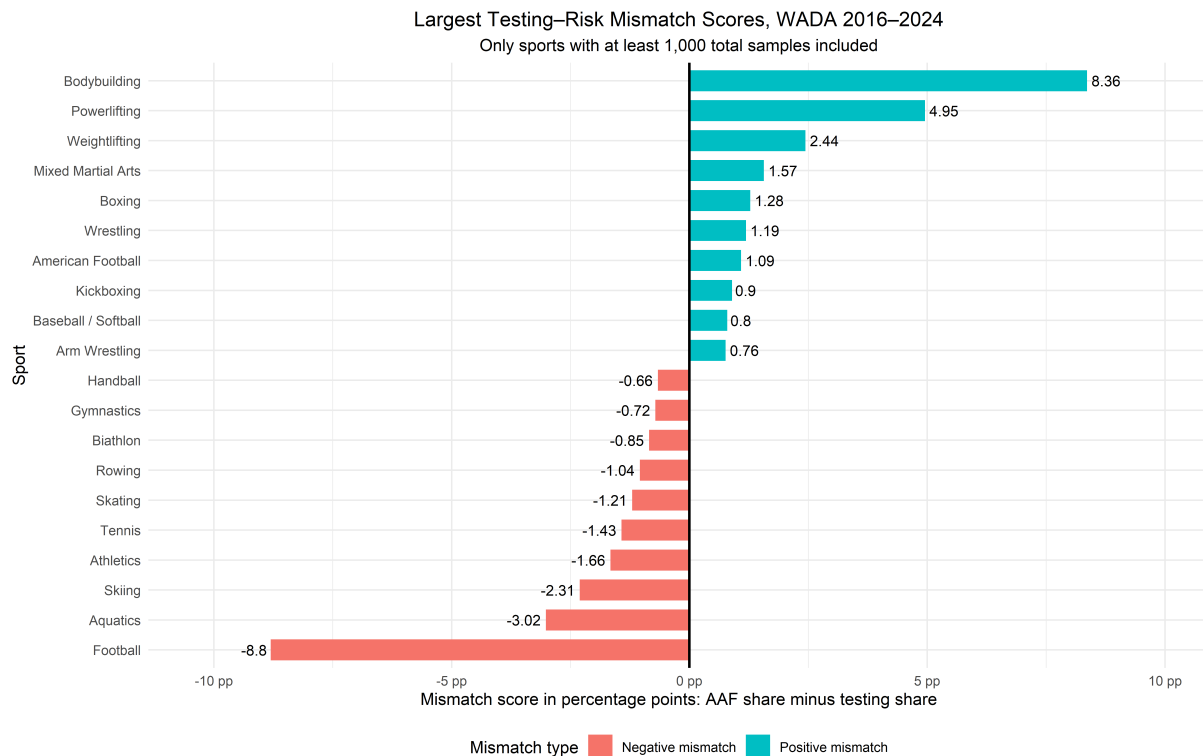
```
## # A tibble: 10 × 7
##   sport_clean total_samples_2016_2...1 total_aafs_2016_2024 testing_share_percent
##   <chr>          <dbl>          <dbl>          <dbl>
## 1 Bodybuildi...    11200         1810           0.5
## 2 Powerlifti...    31168         1295           1.38
## 3 Weightlift...  102298         1427           4.54
## 4 Mixed Mart...   16742          472           0.74
## 5 Boxing          43031          652           1.91
## 6 Wrestling       54893          741           2.44
## 7 American F...    9596          309           0.43
## 8 Kickboxing      5037          229           0.22
## 9 Baseball / ...  11805          270           0.52
## 10 Arm Wrestl...   1973          173           0.09
## # i abbreviated name: 1total_samples_2016_2024
## # i 3 more variables: aaf_share_percent <dbl>,
## # mismatch_score_percentage_points <dbl>, aaf_rate_per_1000_tests <dbl>
```

3.2 Sports with the largest negative mismatch score

A negative mismatch score means that a sport's share of testing is larger than its share of AAFs. These sports are not necessarily unimportant, but from an AAF-yield perspective, they produce fewer AAFs relative to their testing volume.

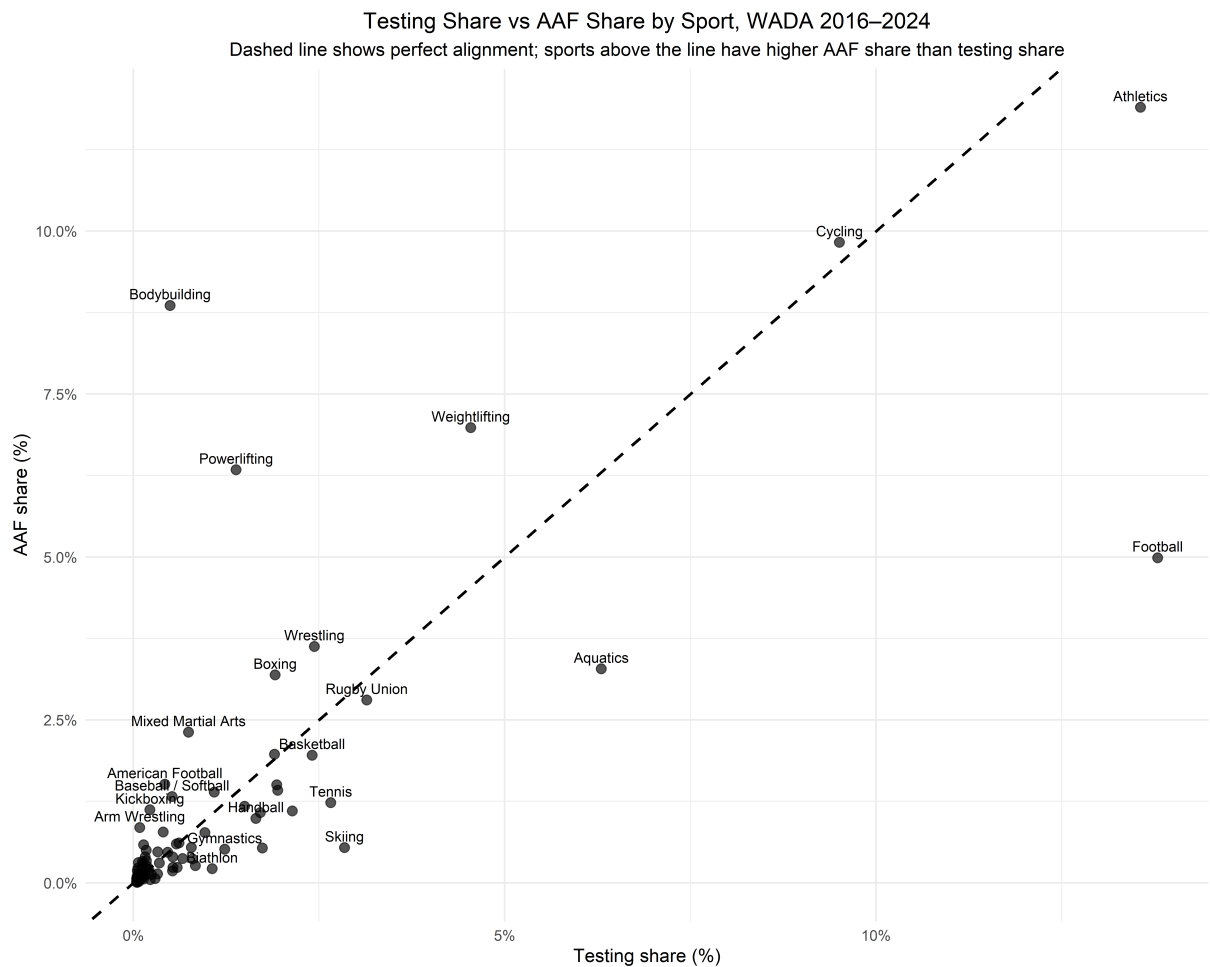
```
## # A tibble: 10 × 7
##   sport_clean total_samples_2016_2...1 total_aafs_2016_2024 testing_share_percent
##   <chr>          <dbl>          <dbl>          <dbl>
## 1 Football       310424         1019          13.8
## 2 Aquatics       141864          671           6.3
## 3 Skiing         64018          110           2.84
## 4 Athletics     305256         2431          13.6
## 5 Tennis         59855          251           2.66
## 6 Skating        39166          109           1.74
## 7 Rowing         48242          225           2.14
## 8 Biathlon       23948           44           1.06
## 9 Gymnastics     27714          105           1.23
## 10 Handball       37163          202           1.65
## # i abbreviated name: 1total_samples_2016_2024
## # i 3 more variables: aaf_share_percent <dbl>,
## # mismatch_score_percentage_points <dbl>, aaf_rate_per_1000_tests <dbl>
```

3.3 Visualising the largest mismatch scores



3.4 Quadrant plot: testing share vs AAF share

This plot compares each sport's testing share with its AAF share. Sports above the diagonal line have a higher AAF share than testing share. Sports below the diagonal line have a lower AAF share than testing share.



3.5 Classifying sports into mismatch categories

The next step classifies sports into four groups. Here, “high” is defined as being in the top 25% of included sports for testing share or AAF share. This gives a relative classification within the dataset.

```
## # A tibble: 4 × 2
##   mismatch_category      n
##   <chr>                <int>
## 1 Lower testing, lower detected risk    59
## 2 High testing, high detected risk     17
## 3 High testing, lower detected risk     5
## 4 Lower testing, high detected risk     5
```

Mismatch category table

mismatch_category	sport	clean_total_samples_2016_2024	total_aafs_2016_2024	testing_share_percent	aaf_share_percent	mismatch_score_percentage
High testing, high detected risk	Football	310424	1019	13.79	4.99	
High testing, high detected risk	Powerlifting	31168	1295	1.38	6.34	
High testing, high detected risk	Aquatics	141864	671	6.30	3.28	
High testing, high detected risk	Weightlifting	102298	1427	4.54	6.98	
High testing, high detected risk	Athletics	305256	2431	13.56	11.90	
High testing, high detected risk	Tennis	59855	251	2.66	1.23	
High testing, high detected risk	Boxing	43031	652	1.91	3.19	
High testing, high detected risk	Wrestling	54893	741	2.44	3.63	
High testing, lower detected risk	Skiing	64018	110	2.84	0.54	
High testing, lower detected risk	Skating	39166	109	1.74	0.53	
High testing, lower detected risk	Gymnastics	27714	105	1.23	0.51	
High testing, lower detected risk	Handball	37163	202	1.65	0.99	

mismatch_category	sport	clean_total_samples_2016_2024	total_aafs_2016_2024	testing_share_percentage	aaf_share_percentage	mismatch_score_percentage
High testing, lower detected risk	Volleyball	38555	220	1.71	1.08	
Lower testing, high detected risk	Bodybuilding	11200	1810	0.50	8.86	
Lower testing, high detected risk	Mixed Martial Arts	16742	472	0.74	2.31	
Lower testing, high detected risk	American Football	9596	309	0.43	1.51	
Lower testing, high detected risk	Kickboxing	5037	229	0.22	1.12	
Lower testing, high detected risk	Baseball / Softball	11805	270	0.52	1.32	
Lower testing, lower detected risk	Biathlon	23948	44	1.06	0.22	
Lower testing, lower detected risk	Arm Wrestling	1973	173	0.09	0.85	
Lower testing, lower detected risk	Badminton	18823	54	0.84	0.26	
Lower testing, lower detected risk	Muaythai	3157	119	0.14	0.58	
Lower testing, lower detected risk	Fencing	17971	75	0.80	0.37	
Lower testing, lower detected risk	Para-Powerlifting	9123	159	0.41	0.78	
Lower testing, lower detected risk	Bobsleigh	13324	48	0.59	0.23	
Lower testing, lower detected risk	Cricket	11913	37	0.53	0.18	

Observation

The mismatch analysis shows that global testing volume and AAF-based detected risk are not perfectly aligned across sports. Some sports account for a much larger share of AAFs than their share of testing, while others receive a large share of testing but produce a smaller share of AAFs.

The strongest positive mismatch appears for Bodybuilding. Although Bodybuilding accounts for only 0.50% of total testing volume, it accounts for 8.86% of all AAFs, producing a mismatch score of +8.36 percentage points. Powerlifting also shows a strong positive mismatch, with 1.38% of tests but 6.34% of AAFs. Other sports with positive mismatch include Weightlifting, Mixed Martial Arts, Boxing, Wrestling, American Football, Kickboxing, Baseball / Softball, and Arm Wrestling.

In contrast, Football shows the largest negative mismatch. It accounts for 13.79% of all tests but only 4.99% of all AAFs, giving a mismatch score of -8.80 percentage points. This means Football receives a very large share of global testing, but its AAF share is much lower than its testing share. Aquatics, Skiing, Athletics, Tennis, Skating, and Rowing also show negative mismatch scores.

Among the most tested sports, the patterns differ. Football and Aquatics show clear negative mismatch, Athletics is closer to alignment but still negative, Cycling is almost aligned, and Weightlifting shows a positive mismatch. This suggests that testing priorities are only partly aligned with AAF-based detected risk.

Overall, RQ3 provides one of the main findings of the project: anti-doping testing is not distributed purely according to AAF yield. Some sports with relatively modest testing volumes, especially Bodybuilding and Powerlifting, produce disproportionately high shares of AAFs, while some very highly tested sports, especially Football, produce lower AAF shares relative to their testing volume.

IMPORTANT FOR RQ3

What we can understand from RQ3: This RQ is more important than RQ1 and RQ2 because it directly asks:

Are testing priorities aligned with detected risk?

The expected key result is:

Pattern	Meaning
Positive mismatch =	Sport has higher AAF share than testing share
Negative mismatch =	Sport has higher testing share than AAF share
Around zero =	Testing share and AAF share are relatively aligned

In the collected data, the most important early finding is that Football is heavily tested but has a much lower AAF share than testing share, while Bodybuilding and Powerlifting show the opposite pattern.

Research Question 4: Do AAF-based risk signals translate into confirmed ADRV outcomes?

The previous section used Adverse Analytical Findings (AAFs) as an indicator of detected doping risk. However, an AAF is not the same as a confirmed anti-doping rule violation. Some AAFs may later be explained by medical reasons, result in no case to answer, lead to no sanction, or remain pending.

Therefore, this section compares AAF-based signals with confirmed Anti-Doping Rule Violations (ADRVs). The aim is to examine whether sports with many AAFs are also the sports with many confirmed violations.

This analysis uses the joined WADA testing and ADRV dataset from **2016 to 2023**, because ADRV data is available up to 2023 in the prepared dataset.

4.1 Overall AAF-to-ADRV outcome summary

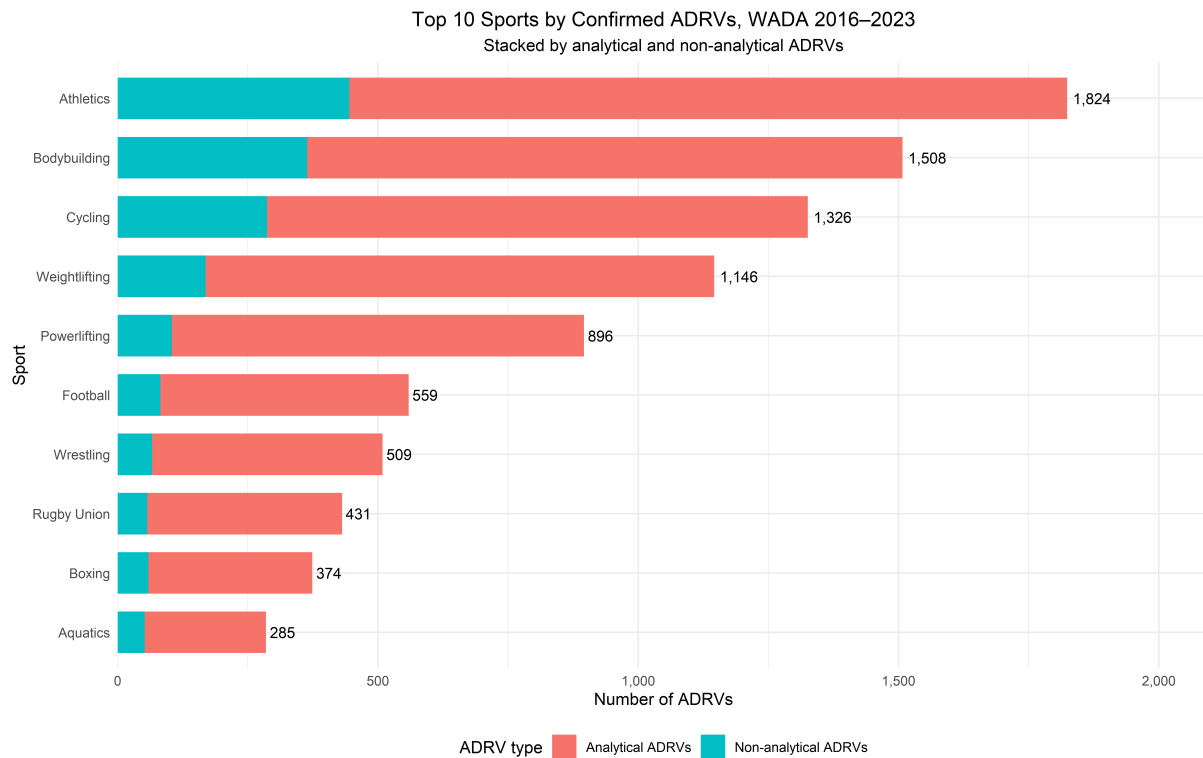
```
## # A tibble: 1 × 9
##   total_samples testing_aafs adrv_report_aaf_outcomes analytical_adrvs
##   <dbl>         <dbl>         <dbl>         <dbl>
## 1     1953134     18122         18096         10756
## # i 5 more variables: non_analytical_adrvs <dbl>, total_adrvs <dbl>,
## #   analytical_adrv_per_testing_aaf <chr>,
## #   analytical_adrv_per_adrv_report_aaf_outcome <chr>,
## #   total_adrv_per_testing_aaf <chr>
```

Overall Table

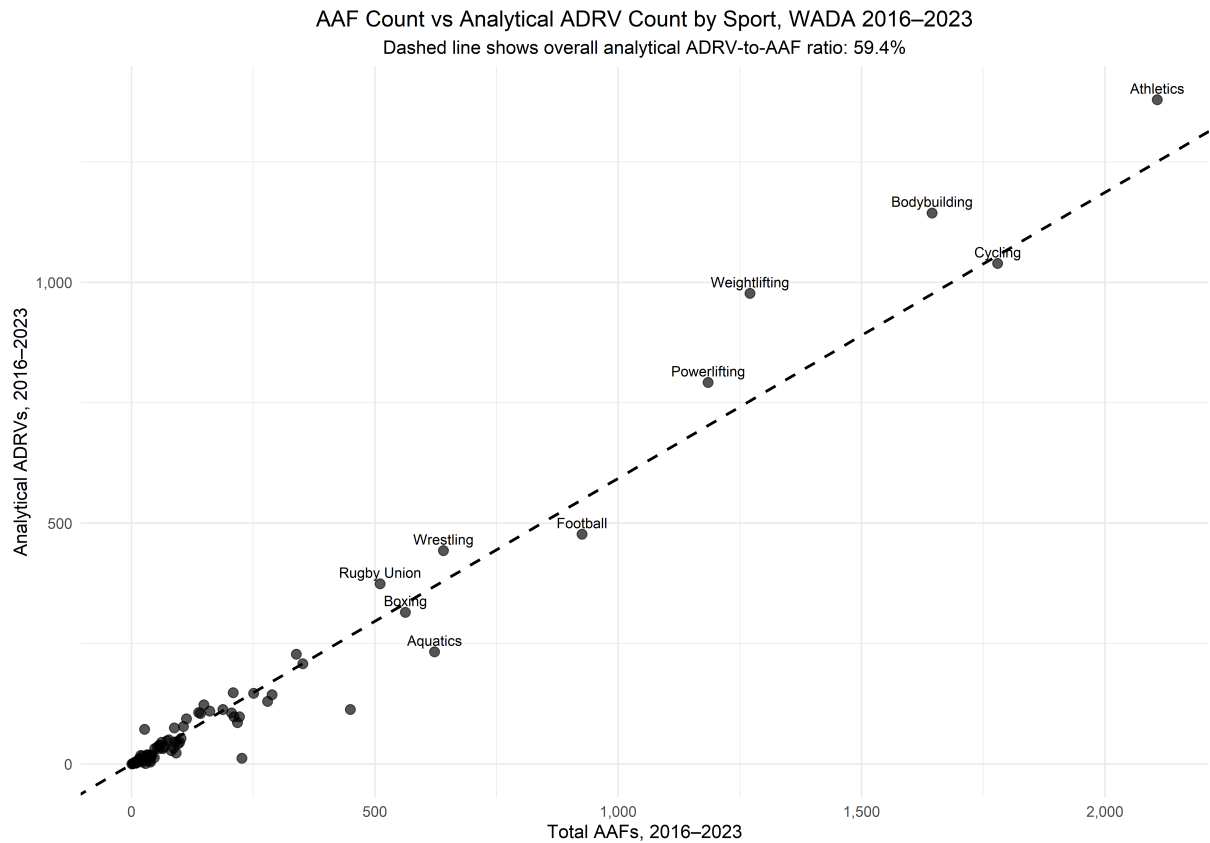
total_sample	testing_aafs	adrv_report_aaf_outcomes	analytical_adrvs	non_analytical_adrvs	total_adrvs	analytical_adrv_per_testing_aaf	analytical_adrv_per_adrv_report_aaf_outcome
1953134	18122	18096	10756	2271	13027	0.5935327	

4.2 Which sports had the most confirmed ADRVs?

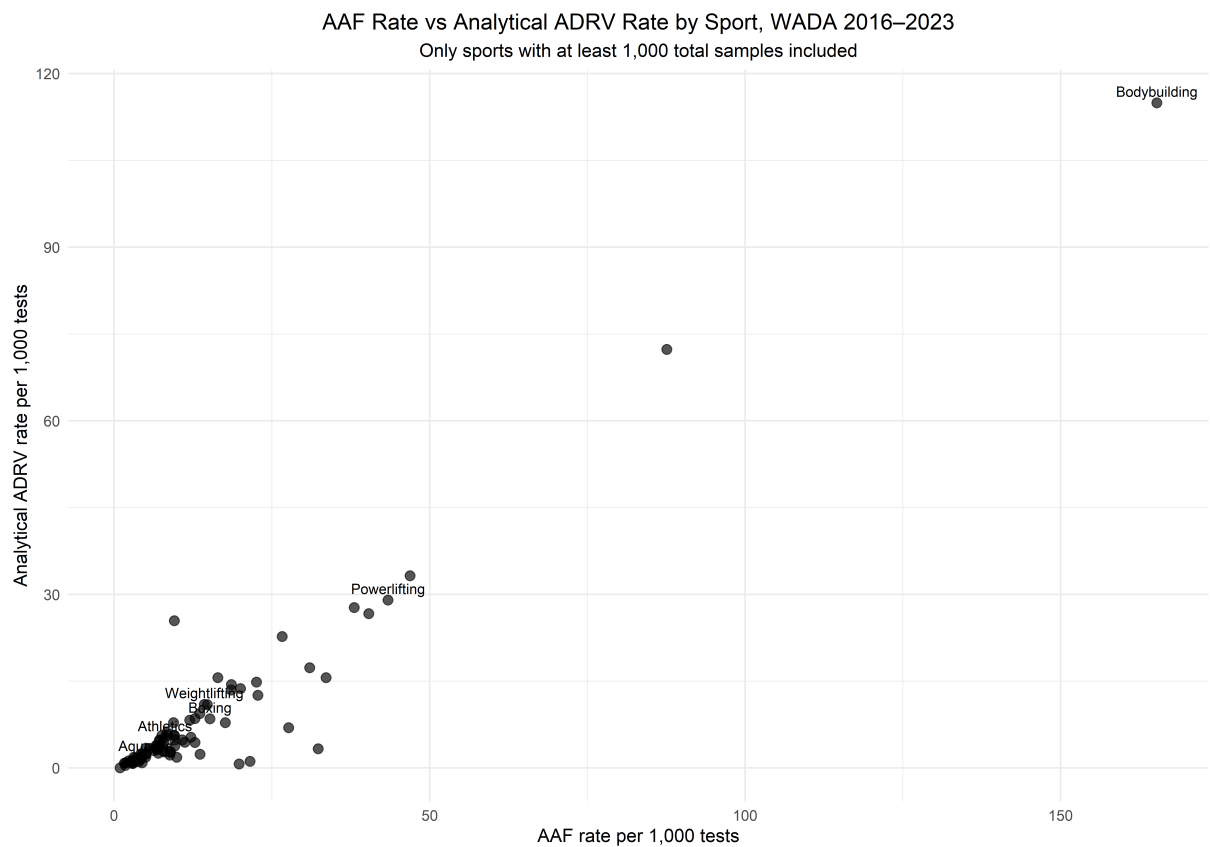
```
## # A tibble: 10 × 7
##   sport_clean total_samples_2016_2023 testing_aafs_2016_2023
##   <chr>         <dbl>         <dbl>
## 1 Athletics           260920           2108
## 2 Bodybuilding          9954           1645
## 3 Cycling            186337           1780
## 4 Weightlifting         88921           1271
## 5 Powerlifting          27295           1185
## 6 Football            273086            926
## 7 Wrestling            47183            641
## 8 Rugby Union          61223            511
## 9 Boxing               37088            563
## 10 Aquatics            124311            623
## # i 4 more variables: analytical_adrvs_2016_2023 <dbl>,
## #   non_analytical_adrvs_2016_2023 <dbl>, total_adrvs_2016_2023 <dbl>,
## #   total_adrv_rate_per_1000_tests <dbl>
```



4.3 Do sports with more AAFs also have more analytical ADRVs?



4.4 Relationship between AAF rate and analytical ADRV rate



Observation

The results show that AAF-based risk signals and confirmed ADRV outcomes are strongly related, but they are not identical. Across 2016–2023, the joined dataset contains 1,953,134 tests, 18,122 testing AAFs, 10,756 analytical ADRVs, and 13,027 total ADRVs. This means that around 59.4% of AAF outcomes became analytical ADRVs.

The sports with the highest number of total ADRVs are Athletics, Bodybuilding, Cycling, Weightlifting, Powerlifting, Football, Wrestling, Rugby Union, Boxing, and Aquatics. Many of these sports also appeared among the highest sports by AAF count, suggesting that strong AAF signals often do translate into confirmed violation outcomes.

However, the relationship is not perfect. Some sports have many AAFs but a lower share of confirmed analytical ADRVs, while others have a relatively high number of ADRVs compared with their AAF count. This is expected because AAFs can have different case outcomes, including medical reasons, no case to answer, no sanction, or pending status. In addition, total ADRVs include non-analytical violations, which do not necessarily originate from laboratory AAFs.

The scatter plot confirms that sports with more AAFs generally also have more analytical ADRVs. Athletics, Bodybuilding, Cycling, Weightlifting, and Powerlifting stand out as sports with both high AAF counts and high confirmed violation counts. This strengthens the conclusion that these sports show consistent detected-risk signals across both AAF and ADRV measures.

Overall, RQ4 shows why it is important not to treat AAFs as confirmed doping violations. AAFs are useful early risk indicators, but confirmed ADRVs provide a stricter outcome-based measure of doping violations.

Research Question 5: Which sports show the strongest confirmed violation risk after adjusting for testing volume?

The previous research question showed that sports with many AAFs often also have many confirmed ADRVs. However, total ADRV counts can be influenced by testing volume. Sports with many tests naturally have more opportunities to detect violations.

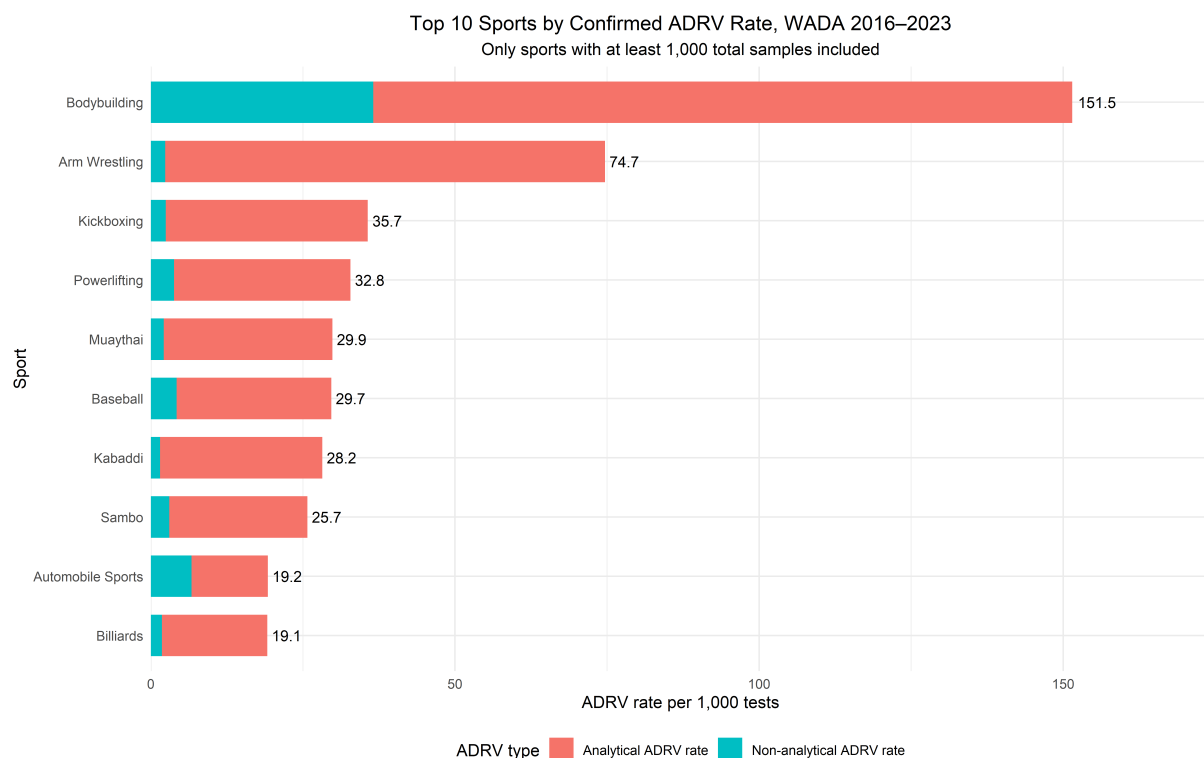
Therefore, this section adjusts confirmed violations by testing volume using the **ADRV rate per 1,000 tests**:

$$\text{ADRV rate per 1,000 tests} = \frac{\text{Total ADRVs}}{\text{Total Samples}} \times 1000$$

This allows comparison between sports with different testing volumes. To avoid unstable rates from very small sports, only sports with at least 1,000 total samples from 2016 to 2023 are included.

5.1 Sports with the highest ADRV rate per 1,000 tests

```
## # A tibble: 10 × 9
##   sport_clean      total_samples_2016_2023 testing_aafs_2016_2023
##   <chr>          <dbl>          <dbl>
## 1 Bodybuilding      9954            1645
## 2 Arm Wrestling     1701             149
## 3 Kickboxing        4457             209
## 4 Powerlifting      27295           1185
## 5 Muaythai          2813             107
## 6 Baseball          2831              27
## 7 Kabaddi           1313              53
## 8 Sambo              3303              88
## 9 Automobile Sports  2547              58
## 10 Billiards        1097              34
## # i 6 more variables: analytical_adrvs_2016_2023 <dbl>,
## #   non_analytical_adrvs_2016_2023 <dbl>, total_adrvs_2016_2023 <dbl>,
## #   total_adrv_rate_per_1000_tests <dbl>,
## #   analytical_adrv_rate_per_1000_tests <dbl>,
## #   non_analytical_adrv_rate_per_1000_tests <dbl>
```

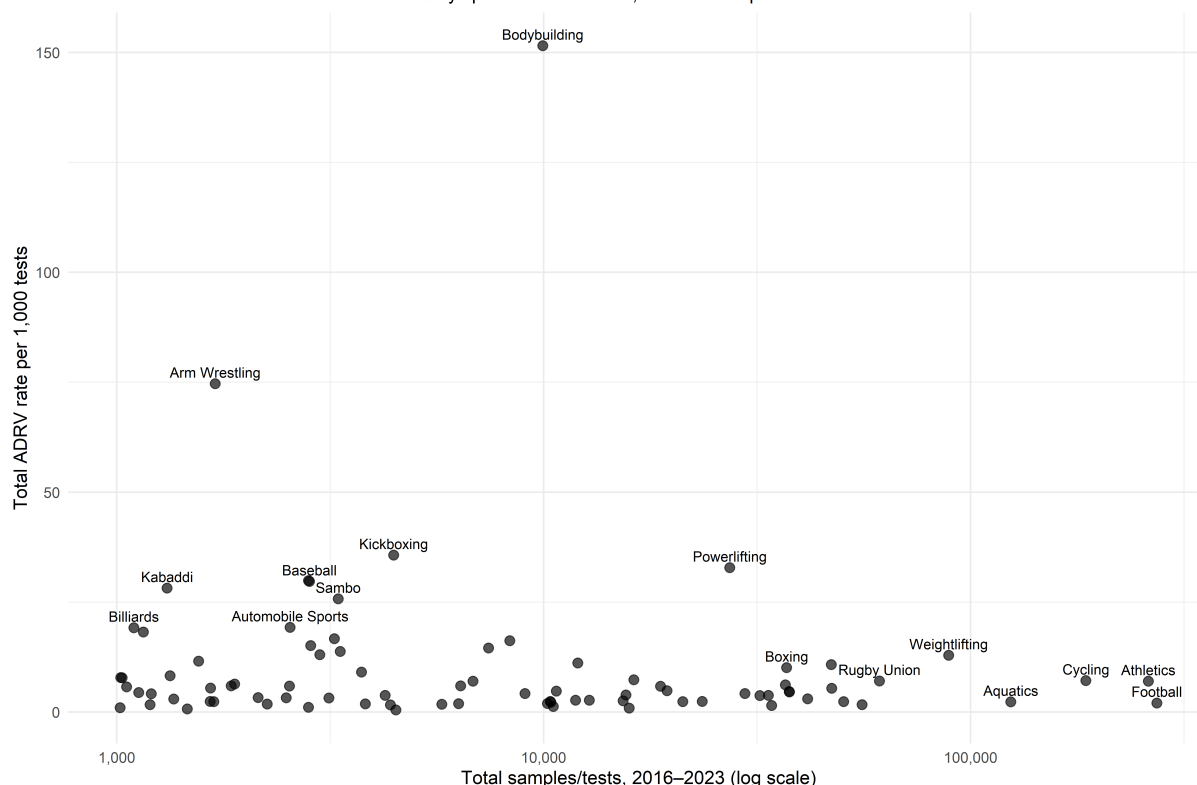


5.2 Testing volume vs confirmed ADRV rate

This plot checks whether highly tested sports are also the sports with the highest confirmed violation rates.

Testing Volume vs Confirmed ADRV Rate by Sport, WADA 2016–2023

Only sports with at least 1,000 total samples included



5.3 Sports with the largest ADRV share compared with testing share

This table compares each sport's share of total testing with its share of total ADRVs. A positive gap means that the sport accounts for a larger share of confirmed ADRVs than its share of tests.

```
## # A tibble: 10 × 8
##   sport_clean      total_samples_2016_2023 total_adrvs_2016_2023
##   <chr>          <dbl>          <dbl>
## 1 Bodybuilding      9954          1508
## 2 Powerlifting      27295          896
## 3 Weightlifting     88921         1146
## 4 Wrestling        47183          509
## 5 Kickboxing       4457           159
## 6 Boxing           37088          374
## 7 Arm Wrestling    1701           127
## 8 Athletics       260920         1824
## 9 Cycling         186337         1326
## 10 American Football 8334           135
## #> 5 more variables: total_adrv_rate_per_1000_tests <dbl>,
## #> testing_share <chr>, adrv_share <chr>,
## #> adrv_testing_gap_percentage_points <dbl>, adrv_to_testing_share_ratio <dbl>
```

Observation

After adjusting confirmed ADRVs by testing volume, the strongest confirmed violation risk is concentrated in a different group of sports than the most-tested sports. Bodybuilding has by far the highest confirmed ADRV rate, with 1,508 total ADRVs from 9,954 tests, equal to 151.5 ADRVs per 1,000 tests. This is much higher than any other sport in the dataset.

Other sports with high confirmed ADRV rates include Arm Wrestling, Kickboxing, Powerlifting, Muaythai, Baseball, Kabaddi, Sambo, Automobile Sports, and Billiards. These sports do not necessarily have the largest total testing volumes, but they produce high numbers of confirmed violations relative to the number of tests conducted.

Among larger and more globally prominent sports, Powerlifting, Weightlifting, Wrestling, Boxing, Cycling, Rugby Union, and Athletics also show important confirmed violation signals. However, their rates are lower than Bodybuilding because their testing volumes are much larger.

The scatter plot shows that high testing volume does not automatically mean high confirmed violation rate. Football and Aquatics receive very large testing volumes, but their ADRV rates per 1,000 tests are relatively low compared with Bodybuilding, Powerlifting, Weightlifting, Wrestling, and Boxing. This confirms that total testing volume and confirmed violation risk are different concepts.

Overall, RQ5 strengthens the main project argument: anti-doping risk should not be interpreted only through total testing volume or total violation count. When confirmed violations are adjusted by testing volume, several lower-volume sports show much stronger confirmed-risk signals than some of the most heavily tested sports.

[For RQ5, the most important chart is 5.1 Top 10 Sports by Confirmed ADRV Rate. The scatter plot is useful as supporting evidence, but the rate chart is the main result.] = For Finja & Franzi

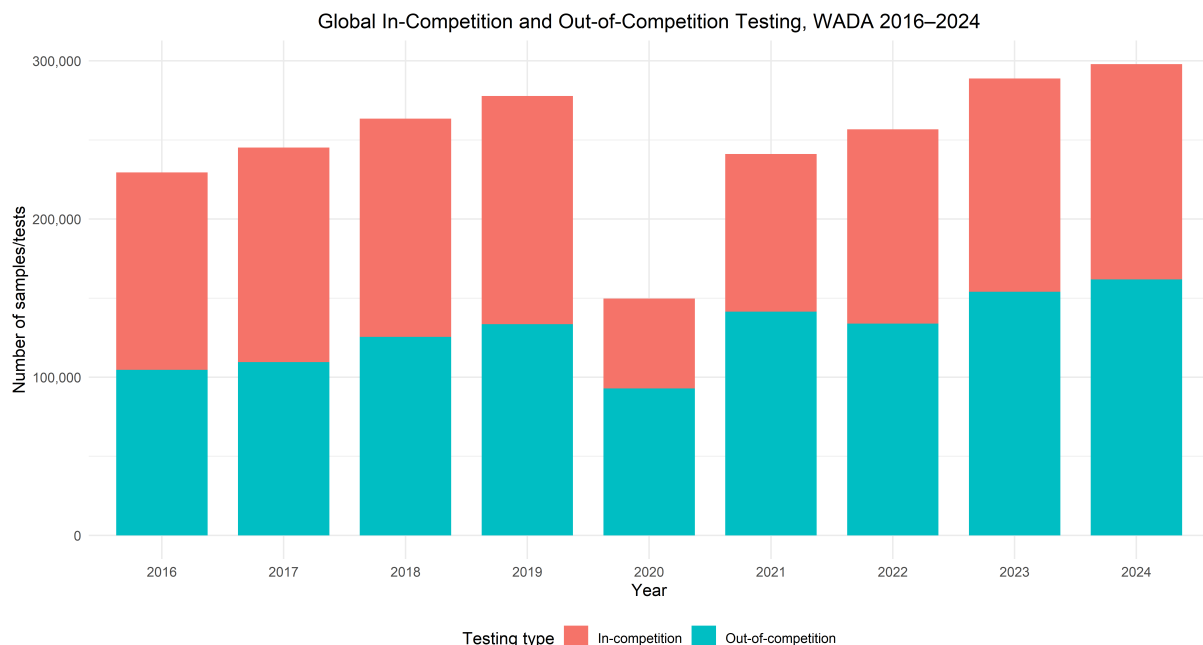
Research Question 6: Does testing strategy differ across higher-risk sports? Or RQ7: How does testing strategy differ across sports: in-competition vs out-of-competition testing?

The previous sections focused on testing volume, AAF-based risk, and confirmed ADRV outcomes. This section examines whether testing strategy differs across sports, especially between **in-competition** and **out-of-competition** testing.

Out-of-competition testing is important because it can detect doping outside the competition period and may be especially relevant for sports where doping risk is not limited to match or event days. Therefore, this section asks whether sports with stronger detected-risk signals also receive higher out-of-competition testing shares.

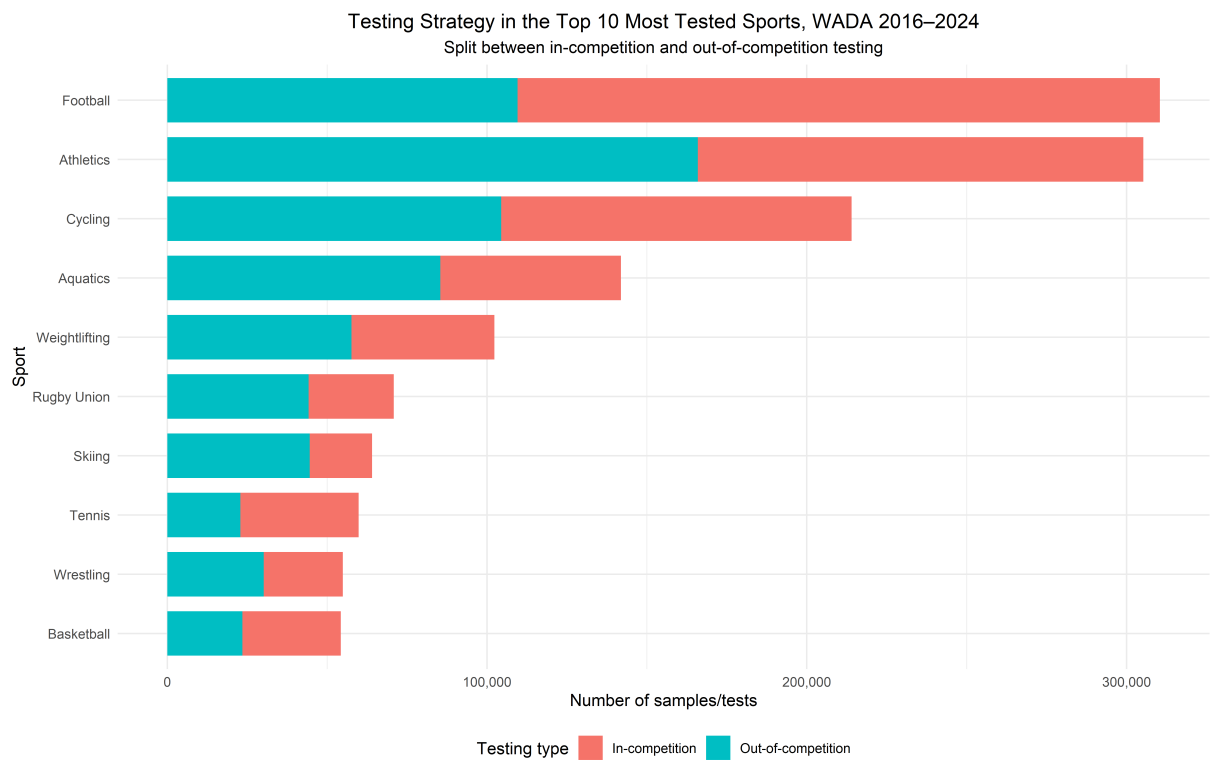
6.1 Global in-competition and out-of-competition testing over time

```
## # A tibble: 9 × 6
##   year total_samples in_competition_samples out_competition_samples
##   <dbl>      <dbl>          <dbl>          <dbl>
## 1 2016      229514          124820          104694
## 2 2017      245232          135694          109538
## 3 2018      263519          138036          125483
## 4 2019      278047          144306          133485
## 5 2020      149758           56881           92859
## 6 2021      241430           99653          141471
## 7 2022      256769          122874          133895
## 8 2023      288865          134805          154060
## 9 2024      297965          136163          161802
## # i 2 more variables: in_competition_share <chr>, out_competition_share <chr>
```



6.2 Testing strategy among the top 10 most tested sports

```
## # A tibble: 10 × 5
##   sport_clean total_samples_2016_2...1 in_competition_share out_competition_share
##   <chr>          <dbl> <chr>          <chr>
## 1 Football      310424 64.7%          35.3%
## 2 Athletics     305256 45.6%          54.4%
## 3 Cycling       214017 51.2%          48.8%
## 4 Aquatics      141864 39.8%          60.2%
## 5 Weightlift... 102298 43.7%          56.3%
## 6 Rugby Union    70806 37.6%          62.4%
## 7 Skiing         64018 30.4%          69.6%
## 8 Tennis         59855 61.8%          38.2%
## 9 Wrestling      54893 45.0%          55.0%
## 10 Basketball    54251 56.8%          43.2%
## # i abbreviated name: 1total_samples_2016_2024
## # i 1 more variable: aaf_rate_per_1000_tests <dbl>
```



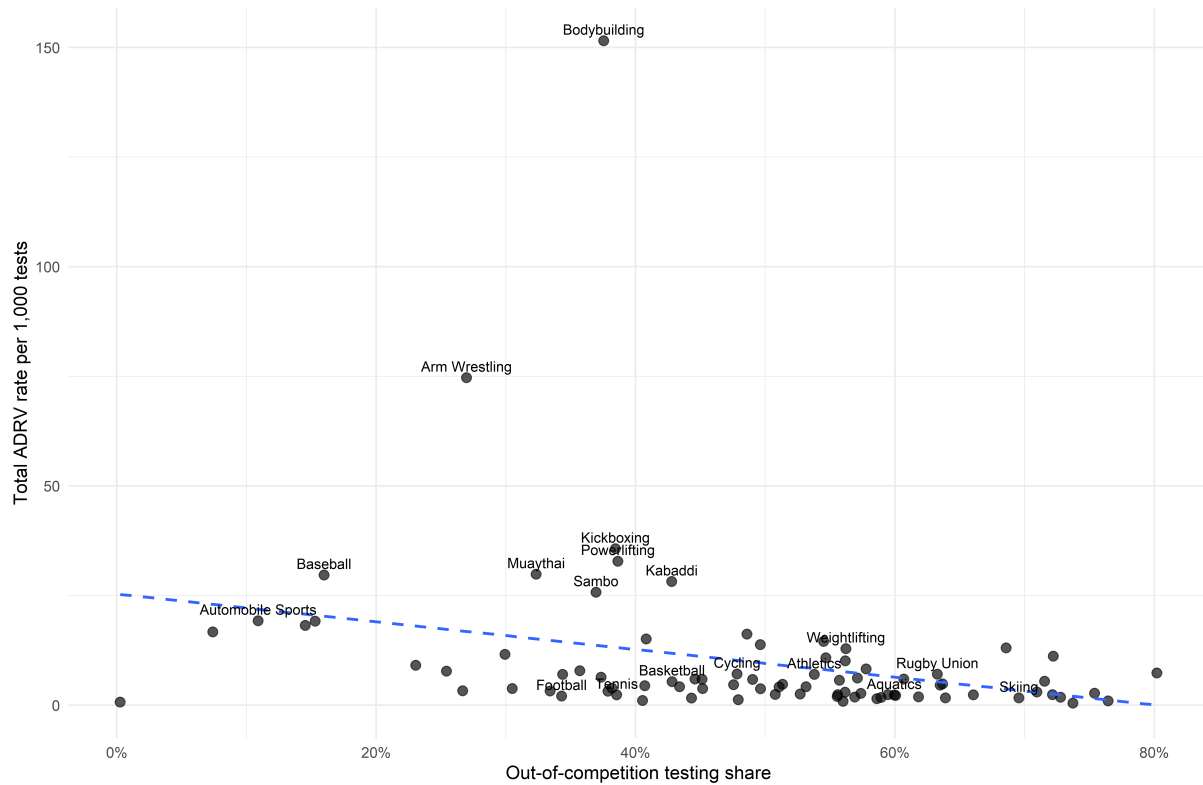
6.3 Out-of-competition testing share and confirmed ADRV risk

Because confirmed ADRV outcomes are available only up to 2023, this part uses the joined dataset from **2016 to 2023**.

```
## # A tibble: 3 × 2
##   relationship                spearman_correlation
##   <chr>                        <dbl>
## 1 OOC share vs AAF rate        -0.45
## 2 OOC share vs analytical ADRV rate -0.44
## 3 OOC share vs total ADRV rate  -0.42
```

```
## # A tibble: 10 × 5
##   sport_clean      total_samples_2016_2023 out_competition_share
##   <chr>                <dbl> <chr>
## 1 Bodybuilding           9954 37.6%
## 2 Arm Wrestling          1701 27.0%
## 3 Kickboxing             4457 38.5%
## 4 Powerlifting           27295 38.7%
## 5 Muaythai               2813 32.3%
## 6 Baseball               2831 16.0%
## 7 Kabaddi                1313 42.8%
## 8 Sambo                  3303 37.0%
## 9 Automobile Sports      2547 10.9%
## 10 Billiards             1097 15.3%
## # i 2 more variables: aaf_rate_per_1000_tests <dbl>,
## #   total_adrv_rate_per_1000_tests <dbl>
```

Out-of-Competition Testing Share vs Confirmed ADRV Rate
Sport-level comparison, WADA 2016–2023



6.4 Comparing high-risk sports with other sports

Here, high-risk sports are defined as the top 25% of included sports by total ADRV rate per 1,000 tests.

```
## # A tibble: 2 × 6
##   adrv_risk_group      number_of_sports average_ooc_share median_ooc_share
##   <chr>                <int> <chr>                <chr>
## 1 High ADRV-rate sports      21 37.8%                38.5%
## 2 Other sports              63 51.8%                53.8%
## # 2 more variables: average_adrv_rate_per_1000 <dbl>,
## #   median_adrv_rate_per_1000 <dbl>
```

Observation

The global testing strategy is almost balanced between in-competition and out-of-competition testing. Across 2016–2024, 48.6% of samples were collected in competition and 51.4% were collected out of competition. However, the balance changed over time. Out-of-competition testing represented 45.6% of testing in 2016, but increased strongly during the COVID-affected years, reaching 62.0% in 2020 and 58.6% in 2021. This suggests that the disruption of regular competitions changed the structure of global anti-doping testing.

Among the top 10 most tested sports, testing strategy differs clearly. Football has a relatively low out-of-competition share, with only 35.3% of tests conducted out of competition. Tennis also has a lower out-of-competition share at 38.2%. In contrast, Skiing, Rugby Union, Aquatics, Weightlifting, Wrestling, and Athletics have higher out-of-competition shares.

When out-of-competition testing share is compared with detected-risk indicators, the relationship is not positive. Among sports with at least 1,000 tests from 2016 to 2023, the Spearman correlation between out-of-competition share and AAF rate is -0.45. The correlation between out-of-competition share and total ADRV rate is -0.42. This means that sports with higher AAF or ADRV rates do not generally have higher out-of-competition testing shares in this dataset.

This pattern is visible among the highest confirmed-risk sports. Bodybuilding, Arm Wrestling, Kickboxing, Powerlifting, Muaythai, Baseball, Kabaddi, Sambo, Automobile Sports, and Billiards have the highest ADRV rates per 1,000 tests, but most of them do not have especially high out-of-competition testing shares. For example, Bodybuilding has an ADRV rate of 151.5 per 1,000 tests, but only 37.6% of its tests are out-of-competition. Powerlifting has an ADRV rate of 32.8 per 1,000 tests, but an out-of-competition share of 38.7%.

Overall, RQ6 suggests that testing strategy is not determined only by detected AAF or ADRV risk. Some heavily tested Olympic or professional sports receive substantial out-of-competition testing even when their ADRV rates are moderate or low, while several high-ADRV-rate sports rely more on in-competition testing. This may reflect differences in sport structure, event calendars, athlete testing pools, and anti-doping priorities beyond simple detected-risk rates. ### RQ6 gives an interesting finding: higher confirmed-risk sports are not necessarily the sports with higher OOC testing shares.

Final Synthesis: Which sports show aligned, over-tested, or under-tested global patterns?

The previous research questions examined testing volume, AAF-based risk, ADRV outcomes, confirmed violation rates, and testing strategy separately. This final synthesis combines these indicators into one sport-level table.

The aim is to identify whether each sport appears to show:

- high testing and high detected risk,

- high testing but lower detected risk,
- lower testing but high detected risk,
- or lower testing and lower detected risk.

Because confirmed ADRV data is available up to 2023, this synthesis uses the joined WADA testing and ADRV dataset from **2016 to 2023**.

Classification rules

The classification uses relative thresholds within the dataset. A sport is defined as **high testing** if its testing share is in the top 25% of included sports. A sport is defined as having high detected risk if its AAF rate or ADRV rate is in the top 25% of included sports.

```
## # A tibble: 4 × 2
##   synthesis_category      n
##   <chr>                <int>
## 1 Lower testing, lower detected risk    40
## 2 Lower testing, high detected risk     23
## 3 High testing, lower detected risk     17
## 4 High testing, high detected risk      4
```

Final synthesis table

The full synthesis table contains all included sports. For readability, the table below shows key sports selected from the most tested sports, highest AAF-rate sports, highest ADRV-rate sports, and strongest mismatch cases.

Final synthesis table of key sports, WADA 2016–2023

Sport	Testing rank	AAF rate rank	ADRV rate rank	Total samples	AAF rate / 1,000	ADRV rate / 1,000	AAF gap pp	ADRV gap pp	VOOC share	Synthesis category	Risk signal	Interpretation
Weightlifting	5	23	1888,921		14.29	12.89	2.60	4.36	56.2%	High testing, high detected risk	High confirmed ADRV risk	Risk share exceeds testing share on both AAF and ADRV measures
Wrestling	10	25	2147,183		13.59	10.79	1.19	1.54	54.7%	High testing, high detected risk	High confirmed ADRV risk	Risk share exceeds testing share on both AAF and ADRV measures
Boxing	14	21	2237,088		15.18	10.08	1.27	1.01	56.2%	High testing, high detected risk	High AAF-only risk	Risk share exceeds testing share on both AAF and ADRV measures
Powerlifting	20	4	427,295		43.41	32.83	5.30	5.60	38.7%	High testing, high detected risk	High AAF + high ADRV risk	Risk share exceeds testing share on both AAF and ADRV measures
Football	1	74	69273,086		3.39	2.05	-8.90	-9.77	3.3%	High testing, lower detected risk	Lower detected risk	Testing share exceeds AAF and ADRV shares
Athletics	2	44	31260,920		8.08	6.99	-1.57	0.76	53.8%	High testing, lower detected risk	Lower detected risk	AAF yield lower than testing share; ADRV signal closer to aligned
Cycling	3	36	28186,337		9.55	7.12	0.44	0.72	47.8%	High testing, lower detected risk	Lower detected risk	Testing and risk shares are broadly aligned
Aquatics	4	62	67124,311		5.01	2.29	-2.91	-4.21	60.0%	High testing, lower detected risk	Lower detected risk	Testing share exceeds AAF and ADRV shares
Rugby Union	6	43	2961,223		8.35	7.04	-0.27	0.20	63.3%	High testing, lower detected risk	Lower detected risk	Testing and risk shares are broadly aligned
Skiing	7	83	7655,743		1.61	1.65	-2.38	-2.17	69.6%	High testing, lower detected risk	Lower detected risk	Testing share exceeds AAF and ADRV shares
Tennis	8	64	6650,469		4.40	2.36	-1.36	-1.68	38.6%	High testing, lower detected risk	Lower detected risk	Testing share exceeds AAF and ADRV shares
Basketball	9	51	4047,329		7.16	5.37	-0.53	-0.46	42.8%	High testing, lower detected risk	Lower detected risk	Testing and risk shares are broadly aligned
Rowing	11	60	5641,561		5.08	2.98	-0.96	-1.18	71.0%	High testing, lower detected risk	Lower detected risk	ADRV yield lower than testing share; AAF signal closer to aligned
Skating	16	79	7834,215		2.81	1.46	-1.23	-1.38	58.6%	High testing, lower detected risk	Lower detected risk	Testing share exceeds AAF and ADRV shares
Volleyball	17	58	5033,629		5.59	3.78	-0.68	-0.75	45.2%	High testing, lower detected risk	Lower detected risk	Testing and risk shares are broadly aligned
Handball	18	61	5232,102		5.02	3.74	-0.75	-0.72	49.6%	High testing, lower detected risk	Lower detected risk	Testing and risk shares are broadly aligned
Mixed Martial Arts	25	10	2716,277		27.65	7.31	1.71	0.09	80.2%	Lower testing, high detected risk	High AAF-only risk	AAF signal stronger than testing share; confirmed signal less strong

Sport	Testing rank	AAF rate rank	ADRV rate rank	Total samples	AAF rate / 1,000	ADRV rate / 1,000	AAF gap pp	ADRV gap pp	VOOC share	Synthesis category	Risk signal	Interpretation
Baseball / Softball	33	14	79	10,536	21.55	1.23	0.74	-0.44	47.9%	Lower testing, high detected risk	High AAF-only risk	Testing and risk shares are broadly aligned
Bodybuilding	37	1	19	954	165.26	151.50	8.81	11.29	37.6%	Lower testing, high detected risk	High AAF + high ADRV risk	Risk share exceeds testing share on both AAF and ADRV measures
American Football	39	7	138	334	33.60	16.20	1.16	0.62	48.6%	Lower testing, high detected risk	High AAF + high ADRV risk	AAF signal stronger than testing share; confirmed signal less strong
Kickboxing	46	3	34	457	46.89	35.67	0.95	1.01	38.5%	Lower testing, high detected risk	High AAF + high ADRV risk	Confirmed ADRV signal stronger than testing share
Sambo	52	11	83	303	26.64	25.73	0.33	0.49	37.0%	Lower testing, high detected risk	High AAF + high ADRV risk	Testing and risk shares are broadly aligned
Baseball	57	37	62	831	9.54	29.67	0.01	0.51	16.0%	Lower testing, high detected risk	High confirmed ADRV risk	Testing and risk shares are broadly aligned
Muaythai	59	6	52	813	38.04	29.86	0.46	0.51	32.3%	Lower testing, high detected risk	High AAF + high ADRV risk	Testing and risk shares are broadly aligned
Automobile Sports	60	12	92	547	22.77	19.24	0.20	0.25	10.9%	Lower testing, high detected risk	High AAF + high ADRV risk	Testing and risk shares are broadly aligned
Arm Wrestling	67	2	21	701	87.60	74.66	0.76	0.91	27.0%	Lower testing, high detected risk	High AAF + high ADRV risk	Testing and risk shares are broadly aligned
Kabaddi	75	5	71	313	40.37	28.18	0.23	0.22	42.8%	Lower testing, high detected risk	High AAF + high ADRV risk	Testing and risk shares are broadly aligned
Goalball	76	8	481	206	32.34	4.15	0.16	-0.02	51.1%	Lower testing, high detected risk	High AAF-only risk	Testing and risk shares are broadly aligned
Billiards	80	9	101	097	30.99	19.14	0.14	0.11	15.3%	Lower testing, high detected risk	High AAF + high ADRV risk	Testing and risk shares are broadly aligned
Biathlon	22	81	652	1,200	1.84	2.36	-0.88	-0.71	66.1%	Lower testing, lower detected risk	Lower detected risk	Testing and risk shares are broadly aligned

Warning: package 'DT' was built under R version 4.3.3

Show 10 entries

Search:

Final synthesis table of key sports, WADA 2016–2023

	Sport	Testing rank	AAF rate rank	ADRV rate rank	Total samples	AAF rate / 1,000	ADRV rate / 1,000	AAF gap pp	ADRV gap pp	VOOC share	Synthesis category
1	Weightlifting	5	23	18	88,921	14.29	12.89	2.6	4.36	56.2%	High testing, high detected risk
2	Wrestling	10	25	21	47,183	13.59	10.79	1.19	1.54	54.7%	High testing, high detected risk
3	Boxing	14	21	22	37,088	15.18	10.08	1.27	1.01	56.2%	High testing, high detected risk
4	Powerlifting	20	4	4	27,295	43.41	32.83	5.3	5.6	38.7%	High testing, high detected risk

5	Football	1	74	69	273,086	3.39	2.05	-8.9	-9.77	34.3%	High testing, lower detected risk
6	Athletics	2	44	31	260,920	8.08	6.99	-1.57	0.76	53.8%	High testing, lower detected risk
7	Cycling	3	36	28	186,337	9.55	7.12	0.44	0.72	47.8%	High testing, lower detected risk
8	Aquatics	4	62	67	124,311	5.01	2.29	-2.91	-4.21	60.0%	High testing, lower detected risk
9	Rugby Union	6	43	29	61,223	8.35	7.04	-0.27	0.2	63.3%	High testing, lower detected risk
10	Skiing	7	83	76	55,743	1.61	1.65	-2.38	-2.17	69.6%	High testing, lower detected risk

Saving the full synthesis table

Observation

The final synthesis shows that global anti-doping testing priorities are only partly aligned with detected doping-risk signals.

Only a small group of sports are classified as **high testing, high detected risk**. In this dataset, Powerlifting, Weightlifting, Wrestling, and Boxing fall into this category. These sports combine relatively high testing exposure with strong AAF or ADRV risk indicators. Their risk share also exceeds their testing share on both AAF and ADRV measures, suggesting that testing attention is supported by detected-risk signals.

A second important group is **lower testing, high detected risk**. This group includes Bodybuilding, Arm Wrestling, Kickboxing, Muaythai, Sambo, Kabaddi, American Football, and Mixed Martial Arts. These sports do not receive the largest testing volumes, but they show high AAF or ADRV rates. Bodybuilding is the clearest example: it has a much lower testing rank than Football, Athletics, or Cycling, but it has the highest AAF and ADRV rate in the dataset. This suggests a possible high-yield or under-prioritised testing area from a detected-risk perspective.

The **high testing, lower detected risk** group includes several globally prominent and heavily tested sports, such as Football, Aquatics, Skiing, Tennis, Rowing, Skating, Basketball, Rugby Union, and Athletics. These sports receive large testing volumes, but their AAF or ADRV rates are not among the highest after adjusting for testing volume. Football is the clearest case: it ranks first by testing volume but has much lower AAF and ADRV shares relative to its testing share.

Some sports are more balanced. Cycling, for example, is highly tested and has large total AAF and ADRV counts, but its testing share and risk shares are broadly closer to alignment than Football. This shows why the synthesis table is useful: a sport can be very important in absolute terms while not appearing extreme in rate-based risk rankings.

Overall, the synthesis supports the main conclusion of Project 1: global anti-doping testing does not follow detected doping risk perfectly. Some highly tested sports show lower detected risk, while several other teams sports show stronger risk signals. Therefore, testing volume, AAF rate, ADRV rate, and mismatch scores need to be interpreted together rather than separately.